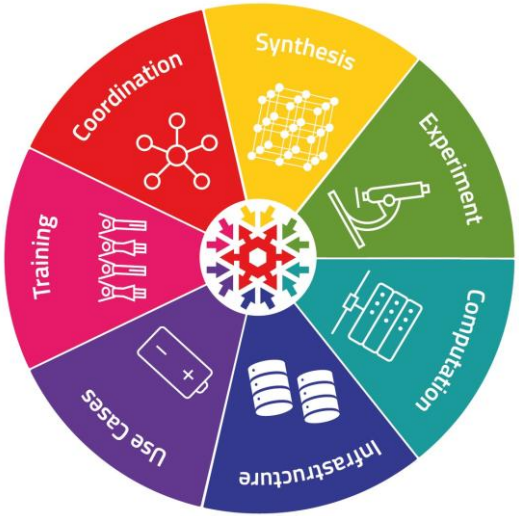
BESSY II - Datathon
March 23, 2026

FAIRmat, NeXus, and NOMAD

Lukas Pielsticker and the FAIRmat team

FAIRmat and NOMAD

FAIRmat is the NFDI consortium to build a FAIR federated data infrastructure for solid-state physics



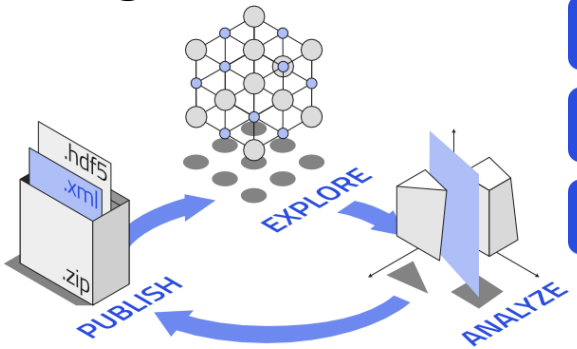
Findable
Accessible
Interoperable
Reusable

NFDI: National (German) Research Data Infrastructure

FAIRmat

NOMAD

FAIRmat hub for **harmonizing** and **reusing** structured data



UPLOADED ENTRIES
16,409,208

UPLOADED FILES
110.5 TB

REGISTERED USERS
> 5000

nomad-lab.eu

NOMAD is a web-based software for FAIR research data management in materials science.

- Processes files to extract structured data and rich metadata.
- Explore and reuse millions of FAIR data from different sources.

The FAIRmat team

~60 PIs + ~30 central coworkers

Claudia Draxl
Spokesperson

Christoph T. Koch
Deputy Spokesperson

FAIRmat uses NeXus for data modelling in experiments

FAIRmat

NeXus is:

- A common data format for experimental science
- A community-driven, open standard
- Widely used at large-scale facilities (neutrons, X-rays, muons) since the 1990s
- Now extended to all of materials science
- Maintained by the international NeXus International Advisory Committee (NIAC)

NX

https://manual.nexusformat.org

NeXus defines:

- A hierarchical structure of groups, fields, and attributes
- *Base classes* (e.g. instrument components like detectors, sources)
- *Application definitions* for specific experiment types
- Conventions for metadata, units, and linking data

NeXus data modelling

- Hierarchical groups representing experiment structure (NXentry, NXinstrument, NXsample, NXdata)
- Fields for numerical data and metadata (with units and physical meaning)
- Attributes to describe context (e.g. units, signal, axes)

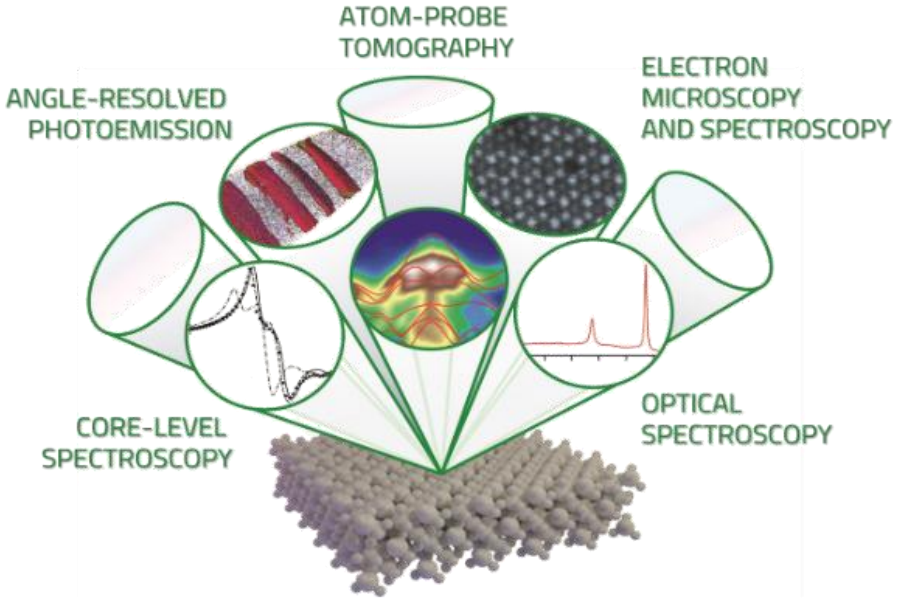
- Typically, serialization of NeXus data is done in HDF5.

```
NXentry
 ├─── definition = "NXdouble_slit"
 ├─── title, start_time, end_time
 ├─── NXinstrument
 │    ├─── source(NXsource)
 │    │    ├─── wavelength  [NX_WAVELENGTH]
 │    │    ├─── coherence_length  [NX_LENGTH, recommended]
 │    │    └─── type  [Laser | Filtered lamp | LED, optional]
 │    ├─── double_slit(NXslit)
 │    │    ├─── x_gap, slit_separation  [NX_LENGTH]
 │    │    └─── height, material  [optional]
 │    └─── detector(NXdetector)
 └─── interference_pattern(NXdata) ← default plot
      ├─── data  dims (n_x, n_y)
      ├─── x_offset  [NX_LENGTH]
      └─── y_offset  [NX_LENGTH]
```

FAIRmat NeXus standardization

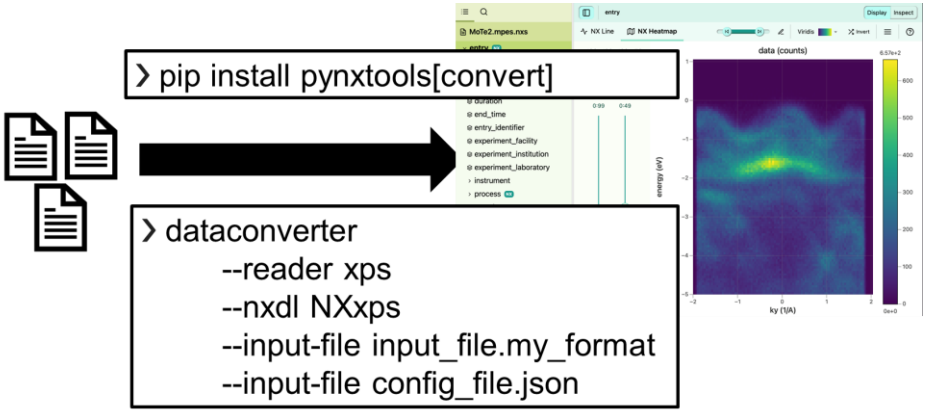
Standardized NeXus (meta)data schema for a subset of methods from materials science

- Application definitions built to be able to represent data from different various source (vendor formats, existing standards, synchrotron data, data analysis software, …)
- **Technology partner workshops** with instrument/data providers and scientists from the community
- FAIRmat contributed
  - 20% of all application definitions
  - 65 % of all base classes
  - 80% of all additional contributions



FAIRmat NeXus tools

- pynxtools
  - converts to NeXus
  - validates and annotates NeXus files



FAIRmat NeXus tools

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- pynxtools plugins
  - provide parsers and mappings
  - typically for one technique
  - plug-in easily to the core pynxtools

Repository	Description
pynxtools-apm	pynxtools reader plugin for atom probe microscopy (APM) as well as related field ion microscopy (FIM) data.
pynxtools-ellips	pynxtools reader plugin for ellipsometry data.
pynxtools-em	pynxtools reader plugin for electron microscopy (EM) data from various vendors/sources.
pynxtools-igor	pynxtools reader plugin for Wavemetrics Igor Pro data.
pynxtools-mpes	pynxtools reader plugin for multi-dimensional photoelectron spectroscopy (MPES) data.
pynxtools-raman	pynxtools reader plugin for Raman data.
pynxtools-spm	pynxtools reader plugin for scanning probe microscopy (SPM) data.
pynxtools-xps	pynxtools reader plugin for X-ray photoelectron spectroscopy (XPS) data from various vendors/sources.
pynxtools-xrd	pynxtools reader plugin for X-ray diffraction data.

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- nyaml
  - XML to YAML converter
  - enables faster NeXus development

```
category: application
doc: |
  This is the most general application definition for
  photoemission experiments.

  Groups and fields are named according to the
  `ISO 18115-1:2023`_ specification as well as the `IUPAC Recommendations 2020`_.

  .. _ISO 18115-1:2023: https://www.iso.org/standard/74811.html
  .. _IUPAC Recommendations 2020: https://doi.org/10.1515/pac-2019-0404
symbols:
  doc: |
    The symbols used in the schema to specify e.g. dimensions of arrays
  n_transmission_function: |
    Number of data points in the transmission function.
type: group
NXmpes(NXobject):
  (NXentry):
    definition:
      \@version:
      enumeration: [NXmpes]
    title:
    start_time(NX_DATE_TIME):
      doc: |
        Datetime of the start of the measurement.
        Should be an ISO8601 date/time stamp. It is recommended to add an explicit time zone,
        otherwise the local time zone is assumed per ISO8601.
    end_time(NX_DATE_TIME):
      exists: recommended
      doc: |
        Datetime of the end of the measurement.
        Should be a ISO8601 date/time stamp. It is recommended to add an explicit time zone,
        otherwise the local time zone is assumed per ISO8601.
    method:
      exists: recommended
      doc: |
        Name of the experimental method.

        If applicable, this name should match the terms given by `Clause 11`_ of
        the `ISO 18115-1:2023`_ specification.

        Examples include:
            * X-ray photoelectron spectroscopy (XPS)
            * angle-resolved X-ray photoelectron spectroscopy (ARXPS)
            * ultraviolet photoelectron spectroscopy (UPS)
            * angle-resolved photoelectron spectroscopy (ARPES)
            * hard X-ray photoemission spectroscopy (HAXPES)
            * near ambient pressure X-ray photoelectron spectroscopy (NAPXPS)
```

NeXus in NOMAD

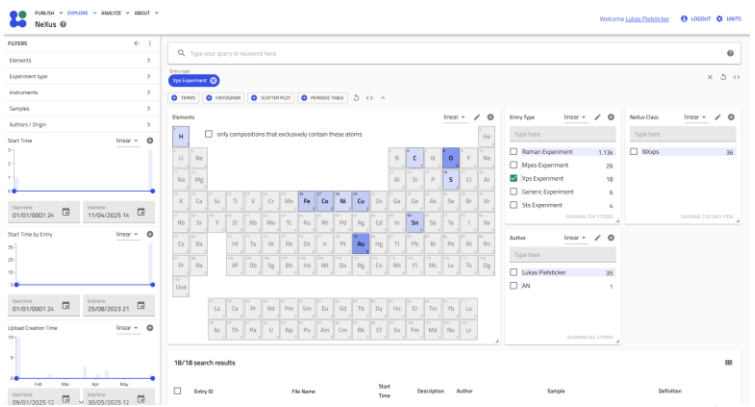
- NeXus is also fully integrated into NOMAD as a NOMAD PLUGIN

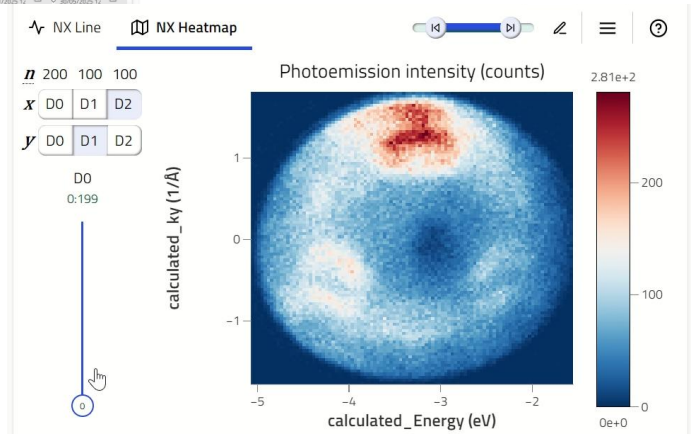
Components:

- NeXus schema
- Parser for NeXus files
- Dataconverter
- visualization (h5web)
- search apps

- You can upload, explore, visualize, and search NeXus data in NOMAD

Have a look: oasis-b





What will you learn in the datathon

FAIRmat

- How to model experiments in NeXus
- How to build a dataconverter pipeline using **pynxtools**
- How to **upload** and **parse** NeXus data in NOMAD
- How to **explore** NeXus data in NOMAD

NOMAD

Schedule

FAIRmat

Day 1: Building an example pipeline

- Setup
- Writing a simple application definition
- Creating a pynxtools reader
- Uploading and visualizing in NOMAD

Day 2: Datathon challenges or bring your own data

https://hz-b.github.io/2026_BESSYII_Datathon/

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